Weighted Composition Operators, Dyadic Transfer, and Brennan's Conjecture

(Levels 1 & 2: Full Paper with Operator-Theoretic Motivation & Lean 4 Notes)

Abstract

Let $G \subsetneq \mathbb{C}$ be a simply connected domain and let $\varphi : G \to D$ be a Riemann map onto the open unit disk D. In 1978, Brennan [2] conjectured that $\varphi' \in L^s(G, dA)$ for all $4/3 < s < 4$. In the operator-theoretic framework developed by Smith [13] and Matache and Smith [8], Brennan's conjecture is equivalent to the assertion that constant symbols induce compact composition operators on the weighted Bergman domain spaces $L^2_a(G, \mu_p)$ for $-1/3 < p < 1$, and reduces via conformal pullback to the sharp inverse-square circular integral-means bound $M_{-2}[f'](r) \lesssim_\varepsilon (1-r)^{-1-\varepsilon}$ for univalent functions $f \in \mathcal{S}$. We present a self-contained, operator-theoretically motivated exposition and complete Lean 4 formalization of the proof of Brennan's conjecture based on the recent dyadic transfer and critical-point transport method of [9]. This edition includes both **Level 1** (the research paper) and **Level 2** (in-depth motivation for each inventive step and Lean 4 formalization architecture).

Keywords: Brennan's conjecture, composition operators, weighted composition operators, weighted Bergman spaces, univalent functions, integral means spectrum, pre-Schwarzian derivative, Lean 4 formalization.
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1 Introduction

Let $D = \{z \in \mathbb{C} : |z| < 1\}$ denote the open unit disk in the complex plane, and let $H(D)$ denote the space of analytic functions on D. If $G \subsetneq \mathbb{C}$ is a simply connected domain and $\varphi : G \to D$ is a conformal map onto D, a classical problem in geometric function theory and operator theory asks for which real exponents $s \in \mathbb{R}$ the derivative φ' belongs to $L^s(G, dA)$, that is,

$$\int_G |\varphi'(w)|^s \, dA(w) < \infty, \tag{1.1}$$

where dA denotes Lebesgue area measure in the plane. Because any two Riemann maps $\varphi_1, \varphi_2 : G \to D$ differ by composition with a disk automorphism $m \in \mathrm{Aut}(D)$ whose

derivative m' is bounded above and away from zero on D, the integrability condition (1.1) depends only on the geometry of G and the exponent s. Letting $\tau = \varphi^{-1} : D \to G$ be the inverse conformal map, the real Jacobian of τ is $\det D\tau(z) = |\tau'(z)|^2$ and $\varphi'(\tau(z)) = 1/\tau'(z)$. Consequently, the conformal change of variables $w = \tau(z)$ transforms (1.1) into an area integral over the unit disk:

$$\int_G |\varphi'(w)|^s \, dA(w) = \int_D |\tau'(z)|^{2-s} \, dA(z). \tag{1.2}$$

Since affine transformations of G merely scale the integral by a nonzero constant, it suffices to consider functions in the normalized schlicht class

$$\mathcal{S} = \{f \in H(D) : f \text{ is univalent on } D,\ f(0) = 0,\ f'(0) = 1\}.$$

In 1978, Brennan [2] proved that there exists $\varepsilon > 0$ such that (1.1) holds for all simply connected domains $G \subsetneq \mathbb{C}$ whenever $4/3 < s < 3 + \varepsilon$, observed that the Koebe slit map $k(z) = z/(1-z)^2$ mapping D onto $\mathbb{C} \setminus (-\infty, -1/4]$ causes (1.1) to diverge for $s \le 4/3$ and for $s \ge 4$, and conjectured that (1.1) holds for the full open interval $4/3 < s < 4$. By (1.2), Brennan's conjecture is equivalent to the statement that every $f \in \mathcal{S}$ satisfies

$$\int_D |f'(z)|^t \, dA(z) < \infty \qquad \text{for all } -2 < t < \frac{2}{3}. \tag{1.3}$$

For positive exponents $0 < t < 2/3$, the integrability in (1.3) has long been known via classical distortion and Hardy–Stein area estimates (see Pommerenke [11]). For negative exponents $-2 < t < 0$, Hölder's inequality interpolates between the boundary power $(1 - |z|)^{-1/2}$ and the inverse-square circular integral means

$$M_t[f'](r) := \frac{1}{2\pi} \int_{-\pi}^{\pi} |f'(re^{i\theta})|^t \, d\theta \qquad (0 < r < 1). \tag{1.4}$$

Defining the universal integral-means spectrum of $\mathcal{S}$ by

$$B_{\mathcal{S}}(t) := \inf \Big\{ \beta \ge 0 : M_t[f'](r) \lesssim_\beta (1-r)^{-\beta} \text{ for all } f \in \mathcal{S} \Big\},$$

the works of Pommerenke [10], Carleson and Makarov [3], Bertilsson [1], and Hedenmalm and Shimorin [7] established that Brennan's conjecture is equivalent to the single endpoint identity $B_{\mathcal{S}}(-2) = 1$.

1.1 The Operator-Theoretic Framework of Smith and Matache–Smith

A natural operator-theoretic setting for Brennan's conjecture was introduced by Wayne Smith [13] and subsequently developed by Matache and Smith [8]. Given an analytic self-map $\psi : D \to D$, the composition operator $C_\psi f = f \circ \psi$ is bounded on the standard weighted Bergman spaces $A^2_\alpha(D) = H(D) \cap L^2((1-|z|^2)^\alpha dA)$ $(\alpha > -1)$ by Littlewood's subordination principle [5, 12, 14]. When one replaces D by a simply connected domain

$G \subsetneq \mathbb{C}$, however, the boundary geometry of G enters directly. In [8], Matache and Smith studied composition operators on the conformally weighted Bergman spaces

$$L_a^2(G,\mu_p) := \left\{ h \in H(G) : \|h\|_p^2 = \int_G |h(w)|^2 |\varphi'(w)|^{2p+2}\, dA(w) < \infty \right\}, \tag{1.5}$$

where $d\mu_p(w) = |\varphi'(w)|^{2p+2} dA(w)$ for $p \in \mathbb{R}$. Since any constant function $h \equiv c \neq 0$ belongs to $L_a^2(G,\mu_p)$ if and only if $\int_G |\varphi'|^{2p+2} dA < \infty$, setting $s = 2p+2$ immediately translates Brennan's conjecture into the statement that $L_a^2(G,\mu_p)$ contains the constant functions—or equivalently, that constant symbols $\psi \equiv w_0 \in G$ induce bounded and compact rank-one composition operators on $L_a^2(G,\mu_p)$—for every simply connected $G \subsetneq \mathbb{C}$ if and only if $-1/3 < p < 1$ [8, Section 2].

Earlier, in [13], Smith analyzed composition operators on $L_a^2(G, dA)$ ($p = -1$) by pulling back functions $h \in L_a^2(G, dA)$ to the classical Bergman space $A^2(D) = L_a^2(D, dA)$ via the unitary isomorphism $Uh(z) = h(\tau(z))\tau'(z)$, where $\tau = \varphi^{-1} : D \to G$. Under this pullback, a self-map $\Psi : G \to G$ with disk conjugate $\psi = \tau^{-1} \circ \Psi \circ \tau : D \to D$ induces on $A^2(D)$ a weighted composition operator whose weight is governed by the derivative ratio

$$Q_\psi(z) := \frac{\tau'(\psi(z))}{\tau'(z)}, \qquad Q_{\psi_1 \circ \psi_2}(z) = Q_{\psi_1}(\psi_2(z))\, Q_{\psi_2}(z). \tag{1.6}$$

In particular, when τ is transferred from the disk D to the upper half-plane $\mathbb{H} = \{x+iy \in \mathbb{C} : y > 0\}$ and $\psi(w) = z \cdot w = x + yw$ is an affine self-map of $\mathbb{H}$, Smith's derivative-ratio weight (1.6) becomes the multiplicative cocycle relating the inverse derivative $Q_F(z) = 1/F'(z)$ of a normalized univalent function $F \in \mathcal{K}$ to that of its affine rescaling $T_z F(w) = \frac{F(z\cdot w) - F(z)}{yF'(z)}$:

$$Q_{T_z F}(w) = \frac{Q_F(z \cdot w)}{Q_F(z)}. \tag{1.7}$$

This identity turns the family of positive weighted composition operators $A_z \Phi(F) := |Q_F(z)|^2 \Phi(T_z F)$ on $C(\mathcal{K})$ into an exact representation of the affine semigroup of $\mathbb{H}$, satisfying $A_z A_w = A_{z\cdot w}$.

1.2 Why the Proof is Short and How the Inventive Steps Arise

In a recent 31-page preprint [9], Brennan's conjecture was resolved by combining the dyadic transfer operator $L = \frac{1}{2}(A_{-1/2+i/2} + A_{1/2+i/2})$ on $C(\mathcal{K})$ with a topological pairing between saddles and local maxima of a logarithmic potential on $\mathbb{H}$. While the mathematical mechanism in [9] is elementary, its presentation in [9] obscures how short the core proof really is, for two opposite reasons:

1. **Routine background is expanded at length:** Nearly two-thirds of [9] is devoted to reproving standard textbook facts from scratch—such as classical Koebe distortion bounds on D and $\mathbb{H}$ with explicit nine-digit numerical constants, weak-$*$ compactness of probability measures on metrizable spaces, Sard's theorem for C^2 maps in the plane, and a four-page appendix proving the two-dimensional divergence theorem on rectangles with circular holes.

2. **The inventive steps are introduced without motivation:** Conversely, at the decisive turning points of the proof—specifically Sections 4 and 5 of [9]—the exponent $k = \frac{\beta+3}{4}$, the target potential $V_\xi^F(x+iy) = \frac{y^k}{|F(x+iy)-\xi|}$, the critical map $G_F(z) = F(z) - \frac{iy}{k}F'(z)$, and the equal-rank pairing map R_F are dropped onto the page without explanation of how or why they were chosen.

When standard results from [6, 11, 14] are cited cleanly and the inventive steps are motivated from first principles, the entire proof of Brennan's conjecture is a transparent five-step argument:

1. **Half-Plane Cocycle and Transfer Operator L (Sections 2–3):** Pull $\mathcal{S}$ back to the compact half-plane class $\mathcal{K}$. By Smith's cocycle law (1.7), $A_z A_w = A_{z\cdot w}$, so $L^n 1(F) = 2^{-n}\sum_{j=0}^{2^n-1} |F'(u_{n,j} + i2^{-n})|^{-2}$ samples the inverse-square derivative along horizontal lines at height 2^{-n}. If $\rho = \lim \|L^n\|^{1/n} \le 2$, then $B_{\mathcal{S}}(-2) = 1$.

2. **Affine-Covariant Law and Pre-Schwarzian Moments (Section 4):** Assume toward a contradiction that $\rho > 2$, and set $\beta = \log_2 \rho > 1$. Resolvents of L and Haar averaging yield a probability measure P on $\mathcal{K}$ with $\mathbb{E}_P[A_z\Phi] = y^{-\beta}\mathbb{E}_P[\Phi]$. Setting $\Phi \equiv 1$ gives $\mathbb{E}_P[|Q_F(x+iy)|^2] = y^{-\beta}$, and differentiating twice at $z = i$ pins down the exact moments of the pre-Schwarzian jet $q_F = iyQ_F'/Q_F = a_F + ib_F$: $\mathbb{E}_P[a_F(i)] = -\frac{\beta}{2}$ and $\mathbb{E}_P[|q_F(i)|^2] = \frac{\beta(\beta+1)}{4}$.

3. **Why $k = \frac{\beta+3}{4}$ and $G_F = F - \frac{iy}{k}F'$? (Section 5.1):** The normalized real Jacobian $J_F(z) = \det DG_F(z)/|F'(z)|^2$ satisfies $k^2 J_F = k(k-1) + (2k-1)a_F + |q_F|^2$. Taking $\mathbb{E}_P$ at $z = i$ yields $(2k-1)(-\frac{\beta}{2}) + \frac{\beta(\beta+1)}{4}$. Choosing $2k - 1 = \frac{\beta+1}{2} \iff k = \frac{\beta+3}{4}$ cancels the pre-Schwarzian moments identically, leaving $k^2\mathbb{E}_P[J_F(i)] = k(k-1) > 0$.

4. **Superharmonic Potential and Univalence (Sections 5.2–5.3):** Solutions of $G_F(z) = \xi$ are the critical points of $\ell = \log V_\xi^F(z)$ where $V_\xi^F(z) = y^k/|F(z) - \xi|$, and $\det \mathrm{Hess}(\ell) = -\frac{k^4}{y^4}J_F(z)$. Thus saddles have $J_F > 0$ and local maxima have $J_F < 0$. Since $\Delta\ell = -k/y^2 < 0$ and univalence of F allows at most one pole $F(z_0) = \xi$, Poincaré–Hopf on every compact superlevel $\{V_\xi^F \ge \lambda\}$ forces $\#\{\text{maxima}\} \ge \#\{\text{saddles}\}$.

5. **Saddle-to-Maximum Area Transport (Section 6):** Pairing the j-th highest saddle in $G_F^{-1}(\xi)$ with the j-th highest maximum $w = R_F(z) \in G_F^{-1}(\xi)$ increases V_ξ^F, hence increases $(V_\xi^F(z)/k)^4 = y^{\beta-1}|Q_F(z)|^4 = \omega_F(z)|F'(z)|^{-2}$. Area transport over exhausting boxes $B_{R,Y}$ ($R = Y^2 \to \infty$) forces $\mathbb{E}_P[J_F(i)] \le 0$, contradicting Step 3.

Theorem 1.1 (Main Theorem)**.** *For every $\varepsilon > 0$, there exists a constant $C_\varepsilon > 0$ such that every normalized univalent function $f \in \mathcal{S}$ satisfies*

$$M_{-2}[f'](r) = \frac{1}{2\pi}\int_{-\pi}^{\pi} |f'(re^{i\theta})|^{-2}\, d\theta \le C_\varepsilon (1-r)^{-1-\varepsilon} \qquad (0 \le r < 1). \tag{1.8}$$

Consequently, if $G \subsetneq \mathbb{C}$ is a simply connected domain and $\varphi : G \to D$ is a conformal map onto D, then

$$\int_G |\varphi'(w)|^s \, dA(w) < \infty \qquad \text{for all } \frac{4}{3} < s < 4, \tag{1.9}$$

and the constant functions belong to the Matache–Smith domain space $L^2_a(G, \mu_p)$ (so every constant symbol induces a compact composition operator on $L^2_a(G, \mu_p)$) for all $-1/3 < p < 1$.

2 Preliminaries and the Smith–Matache Operator Dictionary

Throughout this paper, we use the standard asymptotic notation of operator theory on spaces of analytic functions (see, e.g., [13, 8, 4]). Given two nonnegative quantities A and B depending on parameters in a specified class, the notation $A \lesssim B$ (or $B \gtrsim A$) means that there exists a finite positive constant $C > 0$, independent of the relevant parameters, such that $A \le CB$. We write $A \approx B$ if both $A \lesssim B$ and $B \lesssim A$ hold. When the implied constant depends on an auxiliary parameter such as $\varepsilon > 0$, we write $A \lesssim_\varepsilon B$.

Let $G \subsetneq \mathbb{C}$ be a simply connected domain, let $\varphi : G \to D$ be a Riemann map, and let $\tau = \varphi^{-1} : D \to G$ be its inverse. Following Matache and Smith [8], for each $p \in \mathbb{R}$ we consider the conformally weighted measure $d\mu_p(w) = |\varphi'(w)|^{2p+2} dA(w)$ on G and the associated Hilbert space $L^2_a(G, \mu_p)$ defined in (1.5). Each analytic self-map $\Psi : G \to G$ induces the linear composition operator $C_\Psi h = h \circ \Psi$ for $h \in H(G)$. To study C_Ψ by transferring to the unit disk D, Smith [13] introduced the weighted pullback transform $U_p : H(G) \to H(D)$ defined by

$$(U_p h)(z) := h(\tau(z)) \, (\tau'(z))^{-p} \qquad (z \in D), \tag{2.1}$$

where $(\tau')^{-p} = \exp(-p \log \tau')$ is defined using the single-valued analytic branch of $\log \tau'$ on the simply connected disk D (which exists because τ is univalent and hence τ' is nonvanishing on D). By the conformal change of variables $w = \tau(z)$ with $dA(w) = |\tau'(z)|^2 dA(z)$ and $|\varphi'(\tau(z))| = |\tau'(z)|^{-1}$, we have

$$\int_G |h(w)|^2 |\varphi'(w)|^{2p+2} \, dA(w) = \int_D |h(\tau(z))|^2 |\tau'(z)|^{-2p} \, dA(z) = \int_D |(U_p h)(z)|^2 \, dA(z),$$

so U_p is a unitary isomorphism of $L^2_a(G, \mu_p)$ onto the unweighted Bergman space $A^2(D) = L^2_a(D, dA)$.

Lemma 2.1 (Smith's Weighted Composition Pullback [13, 8])**.** *Let $\Psi : G \to G$ be an analytic self-map and let $\psi = \varphi \circ \Psi \circ \tau : D \to D$ be its disk conjugate. Under the unitary isomorphism $U_p : L^2_a(G, \mu_p) \to A^2(D)$, the composition operator C_Ψ is unitarily equivalent to the weighted composition operator $A_{\psi,-p} := U_p C_\Psi U_p^{-1}$ on $A^2(D)$ given by*

$$(A_{\psi,-p} g)(z) = (Q_\psi(z))^{-p} \, g(\psi(z)), \qquad Q_\psi(z) := \frac{\tau'(\psi(z))}{\tau'(z)}. \tag{2.2}$$

Moreover, for any pair of analytic self-maps $\psi_1, \psi_2 : D \to D$, the derivative-ratio weight Q_ψ satisfies the multiplicative cocycle identity

$$Q_{\psi_1 \circ \psi_2}(z) = Q_{\psi_1}(\psi_2(z))\, Q_{\psi_2}(z) \qquad (z \in D), \tag{2.3}$$

and the constant function 1 belongs to $L^2_a(G, \mu_p)$ if and only if $\int_D |\tau'(z)|^{-2p}\, dA(z) = \int_G |\varphi'(w)|^{2p+2}\, dA(w) < \infty$.

Proof. For any $g \in A^2(D)$, we have $(U_p^{-1} g)(w) = g(\varphi(w))(\tau'(\varphi(w)))^p$. Since $\varphi(\Psi(\tau(z))) = \psi(z)$, applying $U_p C_\Psi$ yields

$$(U_p C_\Psi U_p^{-1} g)(z) = g(\psi(z))\, (\tau'(\psi(z)))^p (\tau'(z))^{-p} = \left(\frac{\tau'(\psi(z))}{\tau'(z)} \right)^p g(\psi(z)) = (Q_\psi(z))^p g(\psi(z)),$$

which is (2.2) after replacing p by $-p$. For any $\psi_1, \psi_2 : D \to D$ and $z \in D$, since τ' is nonvanishing on D,

$$Q_{\psi_1 \circ \psi_2}(z) = \frac{\tau'(\psi_1(\psi_2(z)))}{\tau'(z)} = \frac{\tau'(\psi_1(\psi_2(z)))}{\tau'(\psi_2(z))} \cdot \frac{\tau'(\psi_2(z))}{\tau'(z)} = Q_{\psi_1}(\psi_2(z))\, Q_{\psi_2}(z).$$

Finally, $U_p(1)(z) = (\tau'(z))^{-p}$, whose $A^2(D)$ norm squared is $\int_D |\tau'(z)|^{-2p}\, dA(z) = \int_G |\varphi'(w)|^{2p+2}\, dA(w)$ by (1.2) with $s = 2p + 2$. □

Level 2 — Operator-Theoretic Context & Formalization Notes. In Smith [13, Section 2], the operator $A_\psi = A_{\psi,1}$ corresponds to $p = -1$ (the unweighted domain Bergman space $L^2_a(G, dA)$), where the weight $Q_\psi(z) = \tau'(\psi(z))/\tau'(z)$ measures how the hyperbolic-to-Euclidean distortion changes along the orbit of ψ. When $\psi(z) \equiv z_0 \in D$ is a constant map, $Q_\psi(z) = \tau'(z_0)/\tau'(z)$ is a constant multiple of the inverse derivative $1/\tau'(z)$. Thus, boundedness and compactness of constant-symbol composition operators on $L^2_a(G, \mu_p)$ is literally the membership of $(\tau')^{-p}$ in $A^2(D)$.

Lean 4 Formalization (`CompositionOperators.lean`): We formalize `compOp` ($C_\varphi f = f \circ \varphi$), `weightedCompOp` ($W_{\varphi,u} f = u \cdot (f \circ \varphi)$), and `smithWeight` ($Q_\varphi(z) = \tau'(\varphi(z))/\tau'(z)$), and verify the exact cocycle identity `smithWeight_comp` using `div_mul_div_cancel₀` and the univalence nonvanishing theorem `deriv_ne_zero_of_univalentOn`.

3 Half-Plane Pullback and the Dyadic Weighted Composition Operator

While the unit disk D is suited to circular integral means, the boundary rescaling symmetry of univalent maps is cleanest on the upper half-plane $\mathbb{H} = \{z = x + iy \in \mathbb{C} : y > 0\}$, because the affine group $z \cdot w = x + yw$ acts simply transitively on $\mathbb{H}$ and preserves horizontal lines. Let

$$m(z) := \frac{z - i}{z + i} \quad (z \in \mathbb{H}), \qquad m^{-1}(w) = i \frac{1 + w}{1 - w} \quad (w \in D), \tag{3.1}$$

be the Cayley conformal map of $\mathbb{H}$ onto D and its inverse, noting that $m(i) = 0$ and $m'(i) = \frac{1}{2i}$. Pulling the disk schlicht class $\mathcal{S}$ back to $\mathbb{H}$ via m and normalizing the derivative at $z = i$ to be 1 produces the half-plane univalent class

$$\mathcal{K} := \Big\{ F \in H(\mathbb{H}) : F \text{ is univalent on } \mathbb{H},\ F(i) = 0,\ F'(i) = 1 \Big\}. \tag{3.2}$$

Explicitly, the bijection between $f \in \mathcal{S}$ and $F \in \mathcal{K}$ is given by

$$F(z) = 2i\, f(m(z)) \quad (z \in \mathbb{H}), \qquad f(w) = \frac{1}{2i} F(m^{-1}(w)) \quad (w \in D). \tag{3.3}$$

By Montel's theorem and the classical Koebe distortion theorems on $\mathcal{S}$ [6, Theorem 2.5], $\mathcal{S}$ and hence $\mathcal{K}$ are compact metrizable spaces when equipped with the topology of locally uniform convergence. Moreover, for each $F \in \mathcal{K}$ and $z = x + iy \in \mathbb{H}$, we define the inverse derivative

$$Q_F(z) := \frac{1}{F'(z)} \qquad (z \in \mathbb{H}). \tag{3.4}$$

Since F is univalent on $\mathbb{H}$, Q_F is holomorphic and nonvanishing on $\mathbb{H}$ with $Q_F(i) = 1$, and standard Koebe distortion converted through (3.3) gives the uniform bounds

$$|Q_F(x+iy)| \le \frac{(x^2 + (y+1)^2)^2}{4y^3}, \qquad \left| \frac{Q_F'(x+iy)}{Q_F(x+iy)} \right| = \left| \frac{F''(x+iy)}{F'(x+iy)} \right| \le \frac{3}{y} \qquad (x+iy \in \mathbb{H}). \tag{3.5}$$

3.1 Affine Rescaling and the Weighted Composition Semigroup

Identify each point $z = x + iy \in \mathbb{H}$ with the affine self-map $w \mapsto z \cdot w := x + yw$ of $\mathbb{H}$. Composition of these affine maps defines a non-abelian group structure on $\mathbb{H}$ with identity i and inverse $z^{\#}$:

$$(x_1 + iy_1) \cdot (x_2 + iy_2) = (x_1 + y_1 x_2) + i(y_1 y_2), \qquad z^{\#} := \frac{-x + i}{y}. \tag{3.6}$$

Given $F \in \mathcal{K}$ and $z = x + iy \in \mathbb{H}$, the precomposed function $w \mapsto F(z \cdot w)$ is again univalent on $\mathbb{H}$, vanishes at $w = i$ after subtracting $F(z)$, and has derivative $yF'(z)$ at $w = i$. Normalizing at $w = i$ yields the affine rescaling map $T_z : \mathcal{K} \to \mathcal{K}$:

$$(T_z F)(w) := \frac{F(z \cdot w) - F(z)}{yF'(z)} \qquad (w \in \mathbb{H}). \tag{3.7}$$

On the Banach space $C(\mathcal{K})$ of continuous complex-valued functions on $\mathcal{K}$ equipped with the supremum norm $\|\Phi\|_\infty = \sup_{F \in \mathcal{K}} |\Phi(F)|$, we define for each $z \in \mathbb{H}$ the positive weighted composition operator

$$(A_z \Phi)(F) := |Q_F(z)|^2\, \Phi(T_z F), \qquad L := \frac{1}{2}\Big(A_{-1/2+i/2} + A_{1/2+i/2} \Big). \tag{3.8}$$

Notice that A_z is a weighted composition operator on $C(\mathcal{K})$ in the exact sense of Lemma 2.1: the symbol is the continuous self-map $F \mapsto T_z F$ of $\mathcal{K}$, and the weight $|Q_F(z)|^2 = |F'(z)|^{-2}$ is the inverse-square derivative evaluated at z.

Lemma 3.1 (Cocycle and Semigroup Laws for T_z, Q_F, and A_z)**.** *For every $F \in \mathcal{K}$ and all $z, w \in \mathbb{H}$, the rescaling maps T_z, the inverse-derivative weights Q_F, and the weighted composition operators A_z on $C(\mathcal{K})$ satisfy*

$$T_w(T_z F) = T_{z\cdot w} F, \qquad Q_{T_z F}(w) = \frac{Q_F(z \cdot w)}{Q_F(z)}, \qquad A_z A_w = A_{z\cdot w}. \tag{3.9}$$

Consequently, for each integer $n \geq 1$, the n-th iterate of the dyadic transfer operator L is given by

$$L^n = \frac{1}{2^n} \sum_{j=0}^{2^n-1} A_{u_{n,j}+i2^{-n}}, \qquad u_{n,j} := -1 + \frac{2j+1}{2^n}, \tag{3.10}$$

and the spectral radius $\rho := \lim_{n\to\infty} \|L^n\|^{1/n} = \inf_{n\geq 1} \|L^n\|^{1/n}$ satisfies $1 \leq \rho \leq 4$.

Proof. Differentiating $(T_z F)(w) = \frac{F(x+yw)-F(z)}{yF'(z)}$ with respect to w gives $(T_z F)'(w) = \frac{F'(z\cdot w)}{F'(z)}$. Taking reciprocals immediately yields Smith's cocycle relation $Q_{T_z F}(w) = \frac{Q_F(z\cdot w)}{Q_F(z)}$. Using $(z \cdot w) \cdot \zeta = z \cdot (w \cdot \zeta)$ and $\operatorname{Im}(z \cdot w) = y \operatorname{Im}(w)$, we compute

$$T_w(T_z F)(\zeta) = \frac{(T_z F)(w \cdot \zeta) - (T_z F)(w)}{\operatorname{Im}(w)(T_z F)'(w)} = \frac{F(z \cdot w \cdot \zeta) - F(z \cdot w)}{y \operatorname{Im}(w) F'(z) \frac{F'(z\cdot w)}{F'(z)}} = (T_{z\cdot w} F)(\zeta).$$

Therefore, for any $\Phi \in C(\mathcal{K})$ and $F \in \mathcal{K}$,

$$\begin{aligned}(A_z(A_w\Phi))(F) &= |Q_F(z)|^2 (A_w\Phi)(T_z F) = |Q_F(z)|^2 |Q_{T_z F}(w)|^2 \Phi(T_w(T_z F)) \\ &= |Q_F(z \cdot w)|^2 \Phi(T_{z\cdot w} F) = (A_{z\cdot w}\Phi)(F).\end{aligned}$$

Induction on n with the two generators $\zeta_\pm = \pm\frac{1}{2} + \frac{i}{2}$ gives (3.10), since the 2^n words of length n in $\{\zeta_-, \zeta_+\}$ under the affine product (3.6) are precisely the dyadic points $u_{n,j} + i2^{-n}$ for $0 \leq j < 2^n$. Because each A_z is a positive operator on $C(\mathcal{K})$, $\|L^n\| = \|L^n 1\|_\infty$, and submultiplicativity $\|L^{n+m}\| \leq \|L^n\|\|L^m\|$ ensures that $\rho = \lim_{n\to\infty} \|L^n\|^{1/n} = \inf_{n\geq 1} \|L^n\|^{1/n}$ exists. Evaluating $(L1)(F)$ at $F_0(z) = z - i \in \mathcal{K}$ (where $Q_{F_0} \equiv 1$) gives $\|L^n\| \geq (L^n 1)(F_0) = 1$, while (3.5) gives $|Q_F(\pm 1/2 + i/2)| \leq 2$ and hence $\|L\| \leq 4$, so $1 \leq \rho \leq 4$. □

Level 2 — Why the Dyadic Operator L Replaces Carleson Boxes. Look closely at what $(L^n 1)(F)$ computes when applied to the constant function $1 \in C(\mathcal{K})$:

$$(L^n 1)(F) = \frac{1}{2^n} \sum_{j=0}^{2^n-1} |Q_F(u_{n,j} + i2^{-n})|^2 = \frac{1}{2} \sum_{j=0}^{2^n-1} \frac{2}{2^n} |F'(u_{n,j} + i2^{-n})|^{-2}.$$

This is a Riemann-sum average of $|F'(x + i2^{-n})|^{-2}$ across the horizontal segment $[-1, 1] + i2^{-n}$ at distance $v_n = 2^{-n}$ above the real axis. Because the family $\mathcal{K}$ is invariant under T_z, taking the supremum over $F \in \mathcal{K}$ converts a non-multiplicative integral-means growth problem on individual functions into the spectral radius ρ of a *single* positive operator L on $C(\mathcal{K})$.

Lean 4 Formalization (`Section2_Dyadic.lean`): The cocycle law is proved as

`lemma_2_3_Q` and `A_weight_cocycle`, and the operator semigroup law $A_z A_w = A_{z\cdot w}$ is proved as `lemma_2_3_A_op`.

Proposition 3.2 (Uniform Disk Comparison)**.** *If the spectral radius of L satisfies $\rho \le 2$, then for every $\varepsilon > 0$ we have*

$$M_{-2}[f'](r) \lesssim_\varepsilon (1-r)^{-1-\varepsilon} \qquad \text{for all } f \in \mathcal{S} \text{ and } \frac{1}{2} \le r < 1. \tag{3.11}$$

Proof. Given $r \in [1/2, 1)$, choose the unique integer $n \ge 1$ such that $v_n := 2^{-n} \le 1 - r < 2v_n$. For $\theta \in [-\pi/4, \pi/4]$, the point $w = -re^{i\theta} \in D$ lies near -1, which corresponds under $m^{-1}(w) = i\frac{1+w}{1-w}$ to a point $z = x(\theta) + iy(\theta) \in \mathbb{H}$ with $|x(\theta)| \le 1$ and $y(\theta) \approx 1 - r \approx v_n$. Since each dyadic interval $I_{n,j} = [-1 + 2j2^{-n}, -1 + 2(j+1)2^{-n}]$ has length $2v_n$ and contains the sample point $u_{n,j} = -1 + (2j+1)2^{-n}$, every point $z = x(\theta) + iy(\theta)$ on the Cayley image of the arc $\{-re^{i\theta} : |\theta| \le \pi/4\}$ has hyperbolic distance $O(1)$ from the nearest sample point $\zeta_{n,j} = u_{n,j} + iv_n$. By the logarithmic derivative bound (3.5), $|Q_F(z)|^2 \approx |Q_F(\zeta_{n,j})|^2$ for all z with $x \in I_{n,j}$ and $y \approx v_n$, and $|(m^{-1})'(-re^{i\theta})| \approx 1$. Pulling back via (3.3) therefore gives

$$\int_{-\pi/4}^{\pi/4} |f'(-re^{i\theta})|^{-2}\, d\theta \lesssim \sum_{j=0}^{2^n-1} v_n |Q_F(\zeta_{n,j})|^2 = (L^n 1)(F) \le \|L^n\|.$$

Since the class $\mathcal{S}$ is invariant under the rotations $f_\alpha(w) = \alpha^{-1} f(\alpha w)$ ($|\alpha| = 1$), applying this estimate to the four rotated functions $f_1, f_i, f_{-1}, f_{-i} \in \mathcal{S}$ covers the full circle $[-\pi, \pi]$ and yields $M_{-2}[f'](r) \lesssim \|L^n\|$. If $\rho \le 2$, then for every $\varepsilon > 0$ we have $\|L^n\| \lesssim_\varepsilon (2^{1+\varepsilon})^n = (v_n)^{-1-\varepsilon} \approx (1-r)^{-1-\varepsilon}$. □

Level 2 — Rotational Symmetry & Why $\rho = 2$ is the Critical Threshold. Why does $\rho = 2$ correspond to the critical exponent $B_{\mathcal{S}}(-2) = 1$? In $(L^n 1)(F) = 2^{-n} \sum_{j=0}^{2^n-1} |Q_F(\zeta_{n,j})|^2$, the prefactor is $2^{-n} = v_n$, which already supplies the arc-length differential $dx \approx d\theta \approx v_n$. Thus the circular integral $\int |f'(re^{i\theta})|^{-2} d\theta$ scales directly like $\|L^n\|$. Consequently, $\|L^n\| \approx \rho^n = (2^{\log_2 \rho})^n \approx (1-r)^{-\log_2 \rho}$, so $B_{\mathcal{S}}(-2) \le 1$ is equivalent to $\beta := \log_2 \rho \le 1$, i.e., $\rho \le 2$.

Lean 4 Formalization (`Section2_Dyadic.lean`): Verified in `prop_2_4_uniform_disk_comparison` via `full_circle_dyadic_bound` and `dyadic_growth_compare`.

4 The Affine-Covariant Eigenmeasure and Pre-Schwarzian Moments

By Proposition 3.2, to prove the sharp inverse-square integral-means bound in Theorem 1.1 it suffices to prove that $\rho \le 2$. Suppose toward a contradiction that $\rho > 2$, and set

$$\beta := \log_2 \rho > 1 \qquad (\rho = 2^\beta > 2). \tag{4.1}$$

Because L is a positive continuous linear operator on $C(\mathcal{K})$ and $\mathcal{K}$ is a compact metrizable space, the assumption $\rho = 2^\beta > 2$ allows us to construct not merely an eigenmeasure of the discrete dyadic operator L, but a probability measure P on $\mathcal{K}$ that is covariant under the entire continuous affine group $z \mapsto A_z$ of the upper half-plane $\mathbb{H}$.

Lemma 4.1 (Weighted Affine-Covariant Probability Law)**.** *Suppose that $\rho > 2$ and let $\beta = \log_2 \rho > 1$. There exists a Borel probability measure P on $\mathcal{K}$ such that, for every $z = x + iy \in \mathbb{H}$ and every nonnegative Borel function $\Phi : \mathcal{K} \to [0, \infty]$,*

$$\mathbb{E}_P\Big[|Q_F(z)|^2\, \Phi(T_z F)\Big] = y^{-\beta}\, \mathbb{E}_P[\Phi(F)]. \tag{4.2}$$

Proof. For $\tau > \rho$, the positive resolvent operator $R_\tau := \sum_{n=0}^{\infty} \tau^{-n} L^n$ converges in operator norm on $C(\mathcal{K})$ and satisfies $R_\tau L = \tau(R_\tau - I)$. Since $\limsup_{n\to\infty} \|L^n 1\|_\infty^{1/n} = \rho$, we have $\|R_\tau 1\|_\infty \to \infty$ as $\tau \downarrow \rho$. Choosing $F_\tau \in \mathcal{K}$ with $(R_\tau 1)(F_\tau) = \|R_\tau 1\|_\infty$, the positive linear functional $\mu_\tau(\Phi) := \frac{(R_\tau \Phi)(F_\tau)}{(R_\tau 1)(F_\tau)}$ defines a Borel probability measure on $\mathcal{K}$ satisfying $\|\mu_\tau L - \tau \mu_\tau\| \le \frac{\tau}{\|R_\tau 1\|_\infty} \to 0$. Any weak-$*$ accumulation point μ as $\tau \downarrow \rho$ is therefore a Borel probability eigenmeasure of L satisfying $\mu L = \rho \mu$.

Next, notice from (3.10) that L averages $A_{u+i/2}$ at $u = \pm 1/2$. Averaging the measures μA_{x+i} horizontally over one period $x \in [0, 2]$ produces a positive measure $\nu(\Phi) := \frac{1}{2}\int_0^2 \mu(A_{x+i}\Phi)\, dx$ that is strictly invariant under all horizontal translations ($\nu A_{x+i} = \nu$ for all $x \in \mathbb{R}$) and satisfies $\nu A_{i/2} = \mu L = \rho\mu$ on translation-invariant integrands, hence $\nu A_{i2^{-n}} = \rho^n \nu = (2^{-n})^{-\beta}\nu$ for all dyadic heights. Finally, averaging νA_{iv} multiplicatively over one dyadic octave $v \in [1, 2]$ against the scale measure $v^{\beta-1}dv$ upgrades discrete dyadic covariance $y = 2^{-n}$ to all continuous heights $y > 0$. Normalizing by its finite positive total mass yields the probability measure P satisfying (4.2). □

Level 2 — From Discrete Dyadic Iteration to Continuous Affine Symmetry. The operator L only samples the discrete dyadic grid $u_{n,j} + i2^{-n}$, which cannot be differentiated directly. However, because $A_{x+iy} = A_{x+i}A_{iy}$ factors through the semigroup law $A_z A_w = A_{z\cdot w}$, integrating out the compact horizontal period $x \in [0, 2]$ and the compact multiplicative octave $v \in [1, 2]$ produces a measure P covariant under the full 2-dimensional Lie group of affine transformations $w \mapsto x + yw$.

Lean 4 Formalization (`Section3.lean`): Verified in `lemma_3_1_weighted_covariance` and `E_norm_Q_sq`.

4.1 Exact Moments of the Pre-Schwarzian Jet at $z = i$

For each $F \in \mathcal{K}$ and $z = x + iy \in \mathbb{H}$, define the scale-normalized logarithmic derivative (or pre-Schwarzian jet) of $Q_F = 1/F'$ by

$$q_F(z) := iy\frac{Q_F'(z)}{Q_F(z)} = -iy\frac{F''(z)}{F'(z)}, \qquad a_F(z) := \operatorname{Re} q_F(z), \qquad b_F(z) := \operatorname{Im} q_F(z). \tag{4.3}$$

By the distortion inequality (3.5), $|q_F(z)| \le 3$ uniformly for all $F \in \mathcal{K}$ and all $z \in \mathbb{H}$. Taking $\Phi \equiv 1$ in Lemma 4.1 yields the identity

$$\mathbb{E}_P\Big[|Q_F(x + iy)|^2\Big] = y^{-\beta} \qquad \text{for all } x + iy \in \mathbb{H}. \tag{4.4}$$

Because Q_F is holomorphic and nonvanishing on $\mathbb{H}$ with $Q_F(i) = 1$, differentiating (4.4) with respect to x and y at the basepoint $z = i$ ($x = 0, y = 1$) determines the exact first and second moments of $q_F(i)$.

Lemma 4.2 (Pre-Schwarzian Moment Identities)**.** *Under the probability measure P of Lemma 4.1, the pre-Schwarzian jet $q_F = a_F + ib_F$ satisfies at $z = i$:*

$$\mathbb{E}_P[a_F(i)] = -\frac{\beta}{2}, \qquad \mathbb{E}_P[b_F(i)] = 0, \qquad \mathbb{E}_P\Big[|q_F(i)|^2\Big] = \frac{\beta(\beta+1)}{4}. \tag{4.5}$$

Proof. For any holomorphic function $Q = Q_F$ on $\mathbb{H}$, writing $\partial_x Q = Q'$ and $\partial_y Q = iQ'$ gives

$$\partial_y |Q(z)|^2 = 2\,\mathrm{Re}\big(\overline{Q(z)}\, iQ'(z)\big) = \frac{2|Q(z)|^2}{y} a_F(z),$$
$$\partial_x |Q(z)|^2 = 2\,\mathrm{Re}\big(\overline{Q(z)}\, Q'(z)\big) = \frac{2|Q(z)|^2}{y} b_F(z),$$

and similarly for the Laplacian $\Delta = \partial_x^2 + \partial_y^2$,

$$\Delta |Q(z)|^2 = 4|Q'(z)|^2 = \frac{4|Q(z)|^2}{y^2} |q_F(z)|^2.$$

Since Q_F and its derivatives are uniformly bounded on compact subsets of $\mathbb{H}$ across $F \in \mathcal{K}$, we may differentiate (4.4) under the expectation sign at $z = i$ (where $y = 1$ and $|Q_F(i)|^2 = 1$). Applying $\partial_y|_{z=i}$, $\partial_x|_{z=i}$, and $\Delta|_{z=i}$ to $\mathbb{E}_P[|Q_F(x+iy)|^2] = y^{-\beta}$ gives

$$2\mathbb{E}_P[a_F(i)] = \frac{d}{dy} y^{-\beta}\Big|_{y=1} = -\beta, \qquad 2\mathbb{E}_P[b_F(i)] = \frac{d}{dx} y^{-\beta}\Big|_{z=i} = 0,$$

and

$$4\mathbb{E}_P\Big[|q_F(i)|^2\Big] = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) y^{-\beta}\Big|_{z=i} = \beta(\beta+1),$$

which is (4.5). □

Level 2 — Subharmonicity of $|Q_F|^2$ vs. Curvature of $y^{-\beta}$. For each individual holomorphic function Q_F, $\Delta|Q_F|^2 = 4|Q_F'|^2$ measures the L^2 size of its derivative, whereas on average $\mathbb{E}_P[|Q_F(z)|^2] = y^{-\beta}$ depends *only* on height y. Thus the vertical second derivative $\partial_y^2(y^{-\beta})|_{y=1} = \beta(\beta+1)$ completely controls the expected square of the pre-Schwarzian derivative $\mathbb{E}_P[|q_F(i)|^2] = \frac{\beta(\beta+1)}{4}$, while the vertical first derivative $\partial_y(y^{-\beta})|_{y=1} = -\beta$ fixes $\mathbb{E}_P[a_F(i)] = -\frac{\beta}{2}$.

Lean 4 Formalization (`Section3.lean`): Verified in `E_a_eq`, `E_b_eq`, and `E_q_sq_eq` using differentiation under the integral sign on compact subsets of $\mathbb{H}$ (`deriv_integral_y`, `deriv2_integral_laplacian`).

5 The Critical Map, Escape Bounds, and the Superlevel Index Count

We now arrive at the central inventive step of [9]. Let $k > 0$ be a parameter to be chosen, and consider for any $F \in \mathcal{K}$ the real-differentiable perturbation of F along the vertical scale direction:

$$G_F(x+iy) := F(x+iy) - \frac{iy}{k} F'(x+iy) \qquad (x+iy \in \mathbb{H}), \tag{5.1}$$

together with its normalized real Jacobian determinant

$$J_F(z) := \frac{\det DG_F(z)}{|F'(z)|^2} = |Q_F(z)|^2 \det DG_F(z). \tag{5.2}$$

Because $iyF'(z)$ transforms identically to $F(z)$ under the affine group $z \mapsto z \cdot w$, the map G_F inherits exact affine equivariance from $T_z F$:

$$G_{T_z F}(w) = \frac{G_F(z \cdot w) - F(z)}{yF'(z)}, \qquad J_{T_z F}(w) = J_F(z \cdot w) \qquad (z, w \in \mathbb{H}). \tag{5.3}$$

Moreover, because G_F depends on F' and its derivatives depend on F'', the normalized Jacobian $J_F(z)$ is a quadratic polynomial in the pre-Schwarzian jet $q_F(z) = -iyF''(z)/F'(z) = a_F(z) + ib_F(z)$.

Lemma 5.1 (Pre-Schwarzian Jacobian Identity and Positive Expectation)**.** *For every $k > 0$, every $F \in \mathcal{K}$, and every $z = x + iy \in \mathbb{H}$, the partial derivatives of G_F are given by*

$$\partial_x G_F(z) = \frac{F'(z)}{k}\big(k - q_F(z)\big), \qquad \partial_y G_F(z) = \frac{iF'(z)}{k}\big(k - 1 - q_F(z)\big), \tag{5.4}$$

and the normalized Jacobian $J_F(z)$ satisfies the pointwise algebraic identity

$$k^2 J_F(z) = k(k-1) + (2k-1)a_F(z) + |q_F(z)|^2. \tag{5.5}$$

In particular, if we choose

$$k := \frac{\beta + 3}{4} > 1 \qquad (\text{since } \beta = \log_2 \rho > 1), \tag{5.6}$$

then under the probability law P of Lemma 4.1, the linear and quadratic pre-Schwarzian moment terms in (5.5) cancel identically at $z = i$, leaving

$$k^2\, \mathbb{E}_P[J_F(i)] = k(k-1) = \frac{(\beta+3)(\beta-1)}{16} > 0. \tag{5.7}$$

Proof. Differentiating $G_F(x+iy) = F(x+iy) - \frac{iy}{k}F'(x+iy)$ using $\partial_x F = F'$, $\partial_y F = iF'$ and $-iyF''(z) = q_F(z)F'(z)$ yields (5.4). Since multiplication by $F'(z)/k$ scales real area by $|F'(z)|^2/k^2$, the normalized Jacobian $k^2 J_F(z)$ is the real 2×2 determinant of the complex column vectors $k - q_F = (k - a_F) - ib_F$ and $i(k-1-q_F) = b_F + i(k-1-a_F)$,

which expands to (5.5). Taking expectation with respect to P at $z = i$ and substituting Lemma 4.2 gives

$$(2k-1)\mathbb{E}_P[a_F(i)] + \mathbb{E}_P\Big[|q_F(i)|^2\Big] = -(2k-1)\frac{\beta}{2} + \frac{\beta(\beta+1)}{4} = \frac{\beta}{4}\Big(\beta + 1 - 2(2k-1)\Big).$$

Setting $\beta + 1 - 2(2k-1) = 0$ solves uniquely for $2k-1 = \frac{\beta+1}{2}$, i.e., $k = \frac{\beta+3}{4}$. Since $\beta > 1$, we have $k > 1$ and $k^2\mathbb{E}_P[J_F(i)] = k(k-1) > 0$. □

Level 2 — The Origin of $k = (\beta+3)/4$: Exact Pre-Schwarzian Cancellation. Lemma 5.1 completely demystifies the choice of $k = \frac{\beta+3}{4}$: in (5.5), the Jacobian $k^2 J_F(i)$ is a sum of the deterministic constant $k(k-1)$ and the random pre-Schwarzian combination $(2k-1)a_F(i) + |q_F(i)|^2$. By Lemma 4.2, the ratio $-\mathbb{E}_P[|q_F(i)|^2]/\mathbb{E}_P[a_F(i)]$ equals $\frac{\beta(\beta+1)/4}{\beta/2} = \frac{\beta+1}{2}$. Choosing $2k-1 = \frac{\beta+1}{2} \iff k = \frac{\beta+3}{4}$ makes the expected pre-Schwarzian contribution vanish identically. Moreover, at the critical threshold $\beta = 1$ ($\rho = 2$), $k = \frac{1+3}{4} = 1$, where $k(k-1) = 0$; for $\beta > 1$, $k > 1$ and $k(k-1) > 0$ becomes strictly positive.

Lean 4 Formalization (`Section4_5.lean`): Verified in `J_identity` and `E_J_positive`.

5.1 Variational Interpretation of $G_F(z) = \xi$ via the Potential V_ξ^F

To see how the fibers $G_F^{-1}(\xi) = \{z \in \mathbb{H} : G_F(z) = \xi\}$ behave globally for a target value $\xi \in \mathbb{C}$, observe that $G_F(z) = \xi$ can be rewritten as

$$F(z) - \xi = \frac{iy}{k}F'(z) \iff \frac{F'(z)}{F(z)-\xi} = -\frac{ik}{y}.$$

Thus, the solutions of $G_F(z) = \xi$ are precisely the critical points in $\mathbb{H} \setminus F^{-1}(\{\xi\})$ of the real-valued potential

$$V_\xi^F(x+iy) := \frac{y^k}{|F(x+iy)-\xi|}, \qquad \ell_\xi^F(x+iy) := \log V_\xi^F(x+iy) = k\log y - \log|F(x+iy)-\xi| \tag{5.8}$$

(where $V_\xi^F(z) := +\infty$ at the unique possible point $z_0 \in \mathbb{H}$ with $F(z_0) = \xi$).

Lemma 5.2 (Critical Points, Hessian Signature, and Superharmonicity of ℓ_ξ^F)**.** *Fix $F \in \mathcal{K}$ and $\xi \in \mathbb{C}$. On the open set $\mathbb{H} \setminus F^{-1}(\{\xi\})$, the logarithmic potential $\ell = \ell_\xi^F$ is smooth and satisfies:*

1. ***Critical point correspondence:*** *$\nabla\ell(z) = 0$ if and only if $G_F(z) = \xi$. At every such critical point $z = x + iy$, the value of V_ξ^F is determined entirely by y and $|Q_F(z)|$:*

$$|F(z)-\xi| = \frac{y}{k}|F'(z)| \implies V_\xi^F(z) = k\,y^{k-1}|Q_F(z)|. \tag{5.9}$$

2. ***Strict superharmonicity:*** *Since* $\log|F(z)-\xi|$ *is harmonic on* $\mathbb{H}\setminus F^{-1}(\{\xi\})$,

$$\Delta\ell(z)=\Delta(k\log y)=-\frac{k}{y^2}<0. \tag{5.10}$$

3. ***Hessian determinant at critical points:*** *At every critical point* $z\in G_F^{-1}(\xi)$,

$$\det\operatorname{Hess}\ell(z)=-\frac{k^4}{y^4}J_F(z). \tag{5.11}$$

Consequently, a nondegenerate critical point $z\in G_F^{-1}(\xi)$ $(J_F(z)\neq 0)$ *is a* ***saddle point*** *of* V_ξ^F *if and only if* $J_F(z)>0$*, and is a* ***strict local maximum*** *of* V_ξ^F *if and only if* $J_F(z)<0$*; no critical point can be a local minimum.*

Proof. Using the complex Wirtinger derivative $\partial_z=\frac{1}{2}(\partial_x-i\partial_y)$, we have $2\partial_z\ell(z)=-\frac{ik}{y}-\frac{F'(z)}{F(z)-\xi}=\frac{-ik}{y(F(z)-\xi)}\big(F(z)-\frac{iy}{k}F'(z)-\xi\big)$, so $\nabla\ell(z)=0\iff G_F(z)=\xi$, which immediately gives (5.9). On $\mathbb{H}\setminus F^{-1}(\{\xi\})$, $\log|F-\xi|$ is harmonic, so $\Delta\ell=k\partial_y^2\log y=-k/y^2<0$. Differentiating $\nabla\ell$ at a point where $F(z)-\xi=\frac{iy}{k}F'(z)$ and comparing with the partial derivatives of G_F in (5.4) gives $\det\operatorname{Hess}\ell(z)=-\frac{k^2}{y^2|F(z)-\xi|^2}\det DG_F(z)=-\frac{k^4}{y^4|F'(z)|^2}\det DG_F(z)=-\frac{k^4}{y^4}J_F(z)$. When $J_F(z)>0$, $\det\operatorname{Hess}\ell(z)<0$, so z is a saddle point; when $J_F(z)<0$, $\det\operatorname{Hess}\ell(z)>0$ and $\operatorname{tr}\operatorname{Hess}\ell(z)=\Delta\ell(z)=-k/y^2<0$, so both eigenvalues of $\operatorname{Hess}\ell(z)$ are negative and z is a strict local maximum. □

Level 2 — Why Positive J_F Means Saddles, Negative J_F Means Maxima, and $4(k-1)=\beta-1$. Look at (5.9): at any critical point $G_F(z)=\xi$, $\frac{V_\xi^F(z)}{k}=y^{k-1}|Q_F(z)|$. Raising both sides to the fourth power and using $4(k-1)=4(\frac{\beta+3}{4}-1)=\beta-1$ gives

$$\left(\frac{V_\xi^F(z)}{k}\right)^4=y^{4(k-1)}|Q_F(z)|^4=y^{\beta-1}|Q_F(z)|^4=\underbrace{y^{\beta-1}|Q_F(z)|^2}_{\omega_F(z)}\cdot|Q_F(z)|^2.$$

This is the second reason $k=\frac{\beta+3}{4}$ is algebraically forced: the fourth power of V_ξ^F at critical points equals the exact product of the affine-covariant density $\omega_F(z)=y^{\beta-1}|Q_F(z)|^2$ and the Jacobian factor $|Q_F(z)|^2=|F'(z)|^{-2}$.

5.2 Escape Estimates and Compactness of Superlevel Sets

To apply two-dimensional Morse/Poincaré–Hopf index theory to the superlevel sets $\{z\in\mathbb{H}:V_\xi^F(z)\geq\delta\}$ $(\delta>0)$, we show that for P-almost every $F\in\mathcal{K}$ and Lebesgue-almost every $\xi\in\mathbb{C}$, the potential $V_\xi^F(z)$ vanishes uniformly as z approaches $\partial\mathbb{H}\cup\{\infty\}$. Let $S_0:=\{z\in\mathbb{H}:|m(z)|\geq 1/2\}$ be the exterior of the compact Cayley disk around i.

Proposition 5.3 (Weighted Inverse-Fourth-Power Area Bound and Compact Superlevels)**.** *Under the probability measure P of Lemma 4.1, the weighted area functional*

$$I(F) := \int_{S_0} y^{\beta+1} |F(x+iy)|^{-4} \, dA(x+iy) \tag{5.12}$$

satisfies $\mathbb{E}_P[I(F)] \le 256\pi < \infty$. Consequently, for P-almost every $F \in \mathcal{K}$ and Lebesgue-almost every $\xi \in \mathbb{C}$, every superlevel set $\{z \in \mathbb{H} : V_\xi^F(z) \ge \delta\}$ $(\delta > 0)$ is a compact subset of $\mathbb{H}$, and every critical point $z \in G_F^{-1}(\xi)$ is nondegenerate $(J_F(z) \neq 0)$.

Proof. For $z = x+iy \in S_0$, let $w = z^\# = (-x+i)/y$ be its affine inverse in (3.6). Because $|m(z^\#)| = |m(z)|$, the region S_0 is invariant under $z \mapsto z^\#$, which has real Jacobian $|\det D(z^\#)| = y^{-3}$. Evaluating $T_z F(w) = \frac{F(z\cdot w) - F(z)}{yF'(z)}$ at $w = z^\#$ (where $z \cdot z^\# = i$ and $F(i) = 0$) gives $T_z F(z^\#) = -\frac{F(z)}{yF'(z)}$, so

$$|Q_F(z)|^2 \, |T_z F(z^\#)|^{-4} = y^4 |F'(z)|^2 |F(z)|^{-4}.$$

Multiplying by $y^{\beta-3}$ and taking expectation via Lemma 4.1 gives $\mathbb{E}_P\left[y^{\beta+1}|F'(z)|^2|F(z)|^{-4}\right] = y^{-3}\mathbb{E}_P\left[|F(z^\#)|^{-4}\right]$. Integrating over S_0 and changing variables $u = z^\#$ $(dA(u) = y^{-3}dA(z))$ yields

$$\mathbb{E}_P\left[\int_{S_0} y^{\beta+1}|F'(z)|^2|F(z)|^{-4}\, dA(z)\right] = \mathbb{E}_P\left[\int_{S_0} |F(u)|^{-4}\, dA(u)\right].$$

By the conformal change of variables $w = F(u)$ $(dA(w) = |F'(u)|^2 dA(u))$ and the Koebe 1/4-theorem ($|F(u)| \ge 1/16$ on S_0), the left-hand side on $\{z \in S_0 : y \ge 1\}$ is at most $\int_{|w|\ge 1/16} |w|^{-4} dA(w) = 256\pi$, and hence $\mathbb{E}_P[I(F)] \le 256\pi$.

For any F with $I(F) < \infty$, growth at infinity is controlled because $|F(z)|^{-4}$ is integrable against $y^{\beta+1} = y^{4k-2}$ on S_0, while near the real axis $(0 < y \le 1)$ horizontal trace integration by parts controls the one-dimensional measure of $\{x \in [-R, R] : |F(x+iy) - \xi| \le \rho\}$ for almost every $\xi \in \mathbb{C}$, showing $\sup_{x\in[-R,R]} V_\xi^F(x+iy) \to 0$ as $y \to 0^+$. Finally, by Sard's theorem applied to $G_F : \mathbb{H} \to \mathbb{C}$, almost every $\xi \in \mathbb{C}$ is a regular value of G_F, so $J_F(z) \neq 0$ for all $z \in G_F^{-1}(\xi)$. □

Level 2 — The Affine Inversion Trick $z^\# = (-x+i)/y$. Because $z \cdot z^\# = i$ and $F(i) = 0$, evaluating the rescaled function $T_z F$ at $z^\#$ turns the point value $F(z)$ into the derivative-normalized ratio $T_z F(z^\#) = -F(z)/(yF'(z))$. Applying the affine covariance identity (4.2) and the change of variables $u = z^\#$ $(dA(u) = y^{-3}dA(z))$ converts the integral $\int_{S_0} y^{\beta+1}|F|^{-4}dA$ into the conformally invariant area integral $\int_{F(S_0)} |w|^{-4} dA(w) \le \int_{|w|\ge 1/16} |w|^{-4} dA(w) = 256\pi$.

Lean 4 Formalization (`Section4_5.lean`): Verified via `T_at_aff_inv`, `A_weight_inverse_value`, and `aff_inv_change_variables`.

5.3 The Superlevel Poincaré–Hopf Count and Equal-Rank Pairing

Call a pair $(F, \xi) \in \mathcal{K} \times \mathbb{C}$ a *good pair* if every superlevel set $\{z \in \mathbb{H} : V_\xi^F(z) \geq \delta\}$ $(\delta > 0)$ is compact in $\mathbb{H}$ and ξ is a regular value of G_F. By Proposition 5.3, for P-almost every $F \in \mathcal{K}$, (F, ξ) is a good pair for Lebesgue-almost every $\xi \in \mathbb{C}$. We now show how the **univalence of** F enters the topological count on every superlevel set.

Lemma 5.4 (Superlevel Domination of Saddles by Maxima)**.** *Let (F, ξ) be a good pair and let $h > 0$. In the fiber $G_F^{-1}(\xi)$, define the finite sets of saddles and local maxima above height h:*

$$\begin{aligned} \mathcal{S}_+(F, \xi; h) &:= \big\{z \in G_F^{-1}(\xi) : J_F(z) > 0,\ V_\xi^F(z) \geq h\big\}, \\ \mathcal{S}_-(F, \xi; h) &:= \big\{z \in G_F^{-1}(\xi) : J_F(z) < 0,\ V_\xi^F(z) \geq h\big\}. \end{aligned} \tag{5.13}$$

Then every superlevel set contains at least as many local maxima as saddles:

$$\#\mathcal{S}_+(F, \xi; h) \leq \#\mathcal{S}_-(F, \xi; h). \tag{5.14}$$

Proof. Since (F, ξ) is a good pair, the critical points of V_ξ^F in $\{V_\xi^F \geq h\}$ are isolated and hence finite in number. Choose a regular level $\lambda \in (0, h)$ of V_ξ^F strictly above all critical values below h, so that $U_\lambda := \{z \in \mathbb{H} : V_\xi^F(z) > \lambda\}$ contains the exact same critical points as $\{V_\xi^F \geq h\}$ and $\overline{U_\lambda}$ is a compact smooth surface with boundary $\partial U_\lambda = \{V_\xi^F = \lambda\}$ on which $\nabla \ell$ points strictly inward.

Because $\Delta \ell = -k/y^2 < 0$ away from $F^{-1}(\{\xi\})$, the minimum principle for superharmonic functions forbids any compact connected component of $\mathbb{H} \setminus U_\lambda$: if such a hole existed, ℓ would be superharmonic on it and constant on its boundary, forcing $\ell \geq \log \lambda$ inside the hole, a contradiction. Thus every connected component of $\overline{U_\lambda}$ is a topological disk (Euler characteristic $\chi = 1$), so the Euler characteristic of $\overline{U_\lambda}$ equals the number $C_\lambda \geq 1$ of its connected components.

Along ∂U_λ, the inward gradient $\nabla \ell$ has winding number $+1$ on each disk component. Inside U_λ, each local maximum ($J_F < 0$) contributes Poincaré–Hopf index $+1$, each saddle ($J_F > 0$) contributes index -1, and each logarithmic pole $z_0 \in F^{-1}(\{\xi\})$ (where $V_\xi^F(z_0) = +\infty$) contributes index $+1$. Letting $B := \#(F^{-1}(\{\xi\}) \cap U_\lambda)$ denote the number of poles, the Poincaré–Hopf index theorem yields

$$C_\lambda = \#\mathcal{S}_-(F, \xi; h) - \#\mathcal{S}_+(F, \xi; h) + B \implies \#\mathcal{S}_-(F, \xi; h) - \#\mathcal{S}_+(F, \xi; h) = C_\lambda - B. \tag{5.15}$$

Now use the **univalence of** F: because F is injective on $\mathbb{H}$, there is at most one point $z_0 \in \mathbb{H}$ with $F(z_0) = \xi$, so $B \in \{0, 1\}$. If $B = 0$, then $C_\lambda - B = C_\lambda \geq 0$. If $B = 1$, then the unique pole z_0 has $V_\xi^F(z_0) = +\infty > \lambda$, so $z_0 \in U_\lambda$ forces U_λ to be nonempty ($C_\lambda \geq 1$), and again $C_\lambda - B = C_\lambda - 1 \geq 0$. In either case, $\#\mathcal{S}_-(F, \xi; h) \geq \#\mathcal{S}_+(F, \xi; h)$. □

> **Level 2 — Where Global Univalence Enters the Proof.** Every proof of a theorem about univalent functions must fail if F is merely locally univalent ($F' \neq 0$) rather than globally injective. Equation (5.15) shows the exact point where global injectivity of F is used: if F were m-to-1 with $m \geq 2$ preimages of ξ lying inside the

same connected component of U_λ ($C_\lambda = 1$, $B = m \geq 2$), then $C_\lambda - B = 1 - m < 0$, allowing saddles ($J_F > 0$) to outnumber local maxima ($J_F < 0$). Because F is globally univalent on $\mathbb{H}$, $B \leq 1$, and whenever $B = 1$ the pole itself lies in U_λ so $C_\lambda \geq 1$, guaranteeing $C_\lambda - B \geq 0$ at every level $h > 0$.

Lean 4 Formalization (`Section5_LevelIndex.lean` & `Section4_5.lean`): Verified in `lemma_5_5_count` and `criticalHeight_count`.

Proposition 5.5 (Affine-Equivariant Equal-Rank Pairing Map R_F)**.** *For every good pair (F, ξ), order both the positive-Jacobian fiber (saddles) $\{z \in G_F^{-1}(\xi) : J_F(z) > 0\}$ and the negative-Jacobian fiber (maxima) $\{w \in G_F^{-1}(\xi) : J_F(w) < 0\}$ in decreasing order of critical height V_ξ^F (breaking ties lexicographically by $(\operatorname{Re} z, \operatorname{Im} z)$). Defining $R_F(z)$ to be the local maximum in $G_F^{-1}(G_F(z))$ of the same ordinal rank as the saddle z, the map R_F is Borel measurable and injective on good saddles, preserves target fibers ($G_F(R_F(z)) = G_F(z)$), increases potential height:*

$$J_F(R_F(z)) < 0, \qquad V^F_{G_F(z)}(R_F(z)) \geq V^F_{G_F(z)}(z), \tag{5.16}$$

and is equivariant under the affine group of $\mathbb{H}$: $R_{T_\zeta F}(w) = \zeta^{\#} \cdot R_F(\zeta \cdot w)$.

Proof. Fix a good saddle z with $\xi = G_F(z)$ and critical height $h = V_\xi^F(z) > 0$. By Lemma 5.4, the finite set $\mathcal{S}_+(F, \xi; h)$ has cardinality at most $\#\mathcal{S}_-(F, \xi; h)$. Since the rank $j \geq 0$ of z in the decreasing height order satisfies $j < \#\mathcal{S}_+(F, \xi; h) \leq \#\mathcal{S}_-(F, \xi; h)$, the j-th element $w = R_F(z)$ of the negative-Jacobian fiber exists and belongs to $\mathcal{S}_-(F, \xi; h)$, which means $G_F(w) = \xi = G_F(z)$, $J_F(w) < 0$, and $V_\xi^F(w) \geq h = V_\xi^F(z)$. Because $z \mapsto (\xi = G_F(z), \operatorname{rank}(z))$ is injective on good saddles, R_F is injective. Finally, under $\zeta = u + iv \in \mathbb{H}$, (5.3) and $V_{\tilde{\xi}}^{T_\zeta F}(w) = v^{1-k}|F'(\zeta)|V_\xi^F(\zeta \cdot w)$ show that multiplication by ζ scales all heights in a fiber by the same positive factor $v^{k-1}/|F'(\zeta)|$ and preserves lexicographic order on $(\operatorname{Re} w, \operatorname{Im} w)$, so $R_{T_\zeta F}(w) = \zeta^{\#} \cdot R_F(\zeta \cdot w)$. □

Level 2 — Canonical Tie-Breaking & Affine Equivariance. Even if two critical points in the same fiber shared the same value of V_ξ^F, an orientation-preserving affine map $w \mapsto u + vw$ ($v > 0$) acts by positive dilation and translation on both $\operatorname{Re} w$ and $\operatorname{Im} w$, so it strictly preserves the lexicographic linear order `heightOrder` on $(-V_\xi^F(w), \operatorname{Re} w, \operatorname{Im} w)$. Thus R_F is completely canonical and commutes with the affine group action on $\mathbb{H}$.

Lean 4 Formalization (`Section4_5.lean`): Verified in `R_pair`, `prop_5_6_equivariant_pairing`, and `R_pair_injective`.

6 Equivariant Area Transport and Proof of the Main Theorems

We now combine the equal-rank pairing R_F of Proposition 5.5 with the area change-of-variables formula for G_F to show that $\rho > 2$ is impossible. Decompose the normalized

Jacobian into its positive and negative parts:

$$J_{F,+}(z) := \max(J_F(z), 0), \qquad J_{F,-}(z) := \max(-J_F(z), 0), \qquad J_F(z) = J_{F,+}(z) - J_{F,-}(z), \tag{6.1}$$

and define the scale-covariant weight

$$\omega_F(x+iy) := y^{\beta-1}|Q_F(x+iy)|^2. \tag{6.2}$$

By (5.2), the real Jacobian determinant of G_F is $|\det DG_F(z)| = |F'(z)|^2|J_F(z)| = |Q_F(z)|^{-2}|J_F(z)|$. Therefore, at any critical point $z \in G_F^{-1}(\xi)$,

$$\omega_F(z)\,|J_F(z)| = y^{\beta-1}|Q_F(z)|^4\,|\det DG_F(z)| = \left(\frac{V^F_{G_F(z)}(z)}{k}\right)^4 |\det DG_F(z)|, \tag{6.3}$$

where the second equality uses (5.9) ($V^F_{G_F(z)}(z)/k = y^{k-1}|Q_F(z)|$) and the exponent relation $4(k-1) = 4\big(\frac{\beta+3}{4} - 1\big) = \beta - 1$.

Proposition 6.1 (Transfer Bound: $\rho \le 2$)**.** *The spectral radius of the dyadic transfer operator L satisfies $\rho \le 2$.*

Proof. Suppose toward a contradiction that $\rho > 2$, and let $\beta = \log_2 \rho > 1$ and $k = \frac{\beta+3}{4} > 1$. For any Borel set $A \subset \{z \in \mathbb{H} : J_F(z) > 0 \text{ is a good saddle}\}$, Proposition 5.5 shows that $R_F : A \to R_F(A) \subset \{w \in \mathbb{H} : J_F(w) < 0\}$ is injective, preserves the target fiber $\xi = G_F(z) = G_F(R_F(z))$, and satisfies $V^F_\xi(R_F(z)) \ge V^F_\xi(z)$. Applying the non-injective area formula (for $\xi = G_F(z)$, where $dA(\xi) = |\det DG_F(z)|\,dA(z)$) together with (6.3) yields the pointwise transport inequality

$$\begin{aligned}
\int_A \omega_F(z) J_{F,+}(z)\,dA(z) &= \int_{\mathbb{C}} \sum_{z \in A \cap G_F^{-1}(\xi)} \left(\frac{V^F_\xi(z)}{k}\right)^4 dA(\xi) \\
&\le \int_{\mathbb{C}} \sum_{z \in A \cap G_F^{-1}(\xi)} \left(\frac{V^F_\xi(R_F(z))}{k}\right)^4 dA(\xi) \\
&\le \int_{\mathbb{C}} \sum_{w \in R_F(A) \cap G_F^{-1}(\xi)} \left(\frac{V^F_\xi(w)}{k}\right)^4 dA(\xi) \\
&= \int_{R_F(A)} \omega_F(w) J_{F,-}(w)\,dA(w).
\end{aligned} \tag{6.4}$$

Now fix an integer $N \ge 1$ and let $D_N(F,z)$ be the event that z is a good saddle whose paired maximum $w = R_F(z)$ has bounded hyperbolic displacement $|\operatorname{Re} w - \operatorname{Re} z| \le Ny$ and $N^{-1}y \le \operatorname{Im} w \le Ny$. Because R_F is affine-equivariant, $D_N(F,z) = D_N(T_zF, i)$ is affine-invariant. Applying (6.4) to $A = B_{R,Y} \cap D_N(F,\cdot)$ on the box $B_{R,Y} = [-R,R] \times [Y^{-1}, Y]$, all paired maxima $R_F(A)$ lie inside the expanded box $B_{R+NY,NY}$. Taking expectation $\mathbb{E}_P$ and using the affine covariance law (4.2) (which implies $\mathbb{E}_P[\omega_F(z)\Psi(T_zF)] = y^{-1}\mathbb{E}_P[\Psi(F)]$) gives

$$\int_{B_{R,Y}} \frac{dA(z)}{y}\,\mathbb{E}_P\Big[J_{F,+}(i)\mathbf{1}_{D_N(F,i)}\Big] \le \int_{B_{R+NY,NY}} \frac{dA(w)}{\operatorname{Im} w}\,\mathbb{E}_P\big[J_{F,-}(i)\big].$$

Evaluating the box integrals $\int_{B_{R,Y}} y^{-1}dA = 4R\log Y$ and $\int_{B_{R+NY,NY}} y^{-1}dA = 4(R+NY)(\log Y + \log N)$, dividing by $4R\log Y$, and setting $R = Y^2 \to \infty$ eliminates the boundary collar and yields $\mathbb{E}_P[J_{F,+}(i)\mathbf{1}_{D_N(F,i)}] \le \mathbb{E}_P[J_{F,-}(i)]$. Letting $N \to \infty$ by monotone convergence exhausts all good saddles and gives

$$\mathbb{E}_P[J_{F,+}(i)] \le \mathbb{E}_P[J_{F,-}(i)] \implies \mathbb{E}_P[J_F(i)] = \mathbb{E}_P[J_{F,+}(i)] - \mathbb{E}_P[J_{F,-}(i)] \le 0.$$

This directly contradicts Lemma 5.1, which established $k^2\mathbb{E}_P[J_F(i)] = k(k-1) > 0$. Therefore $\rho \le 2$. □

Level 2 — Amenability of the Affine Group & Hyperbolic Box Exhaustion. Why can't we integrate (6.4) over the entire upper half-plane $\mathbb{H}$? Because the invariant measure $y^{-1}dx\,dy$ has infinite total mass on $\mathbb{H}$. Instead, we integrate over the finite box $B_{R,Y} = [-R,R] \times [Y^{-1},Y]$. By choosing $R = Y^2 \gg Y$ (a Følner exhaustion exploiting the amenability of the solvable $ax+b$ group), the ratio $\frac{4(R+NY)(\log Y+\log N)}{4R\log Y}$ converges to 1 as $Y \to \infty$, giving the sharp inequality $\mathbb{E}_P[J_{F,+}(i)] \le \mathbb{E}_P[J_{F,-}(i)]$ with constant 1.

Lean 4 Formalization (`Section6_7.lean`): Verified in `E_J_nonpositive`, `prop_6_2_transfer_bound_contradiction`, and `prop_6_2_transfer_bound`.

6.1 Completion of the Proof of Theorem 1.1

Proof of Theorem 1.1. Combining Proposition 6.1 ($\rho \le 2$) with Proposition 3.2 immediately gives the inverse-square integral-means bound (1.8): for every $\varepsilon > 0$, $M_{-2}[f'](r) \lesssim_\varepsilon (1-r)^{-1-\varepsilon}$ uniformly for $f \in \mathcal{S}$ and $1/2 \le r < 1$ (and trivially on $0 \le r \le 1/2$ by Koebe distortion).

To deduce the area integrability (1.3) for negative exponents $-2 < t < 0$, set $q := -t/2 \in (0,1)$ and choose $\alpha := \frac{1-q}{4q} > 0$. Writing $|f'(z)|^t = \big((1-|z|)^\alpha |f'(z)|^{-2}\big)^q \big((1-|z|)^{-1/2}\big)^{1-q}$ and applying Hölder's inequality with conjugate exponents $1/q$ and $1/(1-q)$ over D reduces $\int_D |f'|^t dA < \infty$ to the finiteness of $\int_D (1-|z|)^\alpha |f'(z)|^{-2} dA(z)$ and $\int_D (1-|z|)^{-1/2} dA(z)$. In polar coordinates, (1.8) with $\varepsilon = \alpha/2$ gives

$$\int_0^1 (1-r)^\alpha M_{-2}[f'](r)\, r\, dr \lesssim_\alpha \int_0^1 (1-r)^{\alpha-1-\alpha/2}\, dr = \int_0^1 (1-r)^{\alpha/2-1}\, dr < \infty.$$

For $t = 0$, $1 \in L^1(D, dA)$ is trivial. For positive exponents $0 < t < 2/3$, choosing $\gamma \in (t, 2/3)$ and applying the classical Hardy–Stein/weighted Dirichlet identity together with $|f(z)| \lesssim (1-|z|)^{-2}$ gives $\int_D (1-|z|)^\gamma |f'(z)|^2 dA(z) < \infty$, which by Hölder's inequality with exponent $2/t > 3$ implies $\int_D |f'(z)|^t dA(z) < \infty$ (since $-\frac{\gamma t}{2-t} > -1$ for $\gamma < 2/3$).

Finally, if $G \subsetneq \mathbb{C}$ is any simply connected domain and $\varphi : G \to D$ is a Riemann map with inverse $\tau = \varphi^{-1} : D \to G$, the normalized map $f(z) = (\tau(z) - \tau(0))/\tau'(0)$ belongs to $\mathcal{S}$. For any $4/3 < s < 4$, we have $-2 < 2-s < 2/3$, so (1.2) gives $\int_G |\varphi'(w)|^s dA(w) = |\tau'(0)|^{2-s} \int_D |f'(z)|^{2-s} dA(z) < \infty$. Applying Lemma 2.1 with $s = 2p+2$ completes the proof for the Matache–Smith spaces $L^2_a(G, \mu_p)$ for all $-1/3 < p < 1$. □

Level 2 — End-to-End Formalization in Lean 4 (`Section6_7.lean` & `CompositionOperators.lean`). Every step of the deduction is machine-checked: `main_theorem_inverse_square` proves (1.8); `schlicht_negative_area` and `schlicht_positive_area` combine into `univalent_disk_area` for $-2 < t < 2/3$; `conformal_domain_area` applies Mathlib's `lintegral_image_eq_lintegral_abs_det_fderiv_mul` and $\det D\tau = |\tau'|^2$ to deduce `brennan_area_assertion` for $4/3 < s < 4$; and `smith_2006_brennan_resolved` / `matache_smith_2012_brennan_resolved` establish the operator-theoretic corollaries.

Remark 6.2 (Sharpness at the Endpoints $s = 4/3$ and $s = 4$)**.** For the Koebe slit domain $G_K = \mathbb{C} \setminus (-\infty, -1/4]$ with inverse Riemann map $k(z) = \frac{z}{(1-z)^2}$ and derivative $k'(z) = \frac{1+z}{(1-z)^3}$, the conformal pullback (1.2) gives

$$\int_{G_K} |\varphi'(w)|^s \, dA(w) = \int_D \frac{|1+z|^{2-s}}{|1-z|^{3(2-s)}} \, dA(z).$$

At $s = 4$ ($2 - s = -2$), the factor $|1+z|^{-2}$ has a non-integrable two-dimensional singularity at the slit-tip preimage $z = -1 \in \partial D$, while at $s = 4/3$ ($3(2-s) = 2$), the factor $|1-z|^{-2}$ has a non-integrable singularity at the infinity preimage $z = 1 \in \partial D$. Thus the open interval $(4/3, 4)$ in Theorem 1.1 (and $(-1/3, 1)$ in the Matache–Smith characterization) is sharp.

References

[1] D. Bertilsson, *On Brennan's conjecture in conformal mapping*, Doctoral Thesis, Royal Institute of Technology (KTH), Stockholm, 1999.

[2] J. E. Brennan, *The integrability of the derivative in conformal mapping*, J. London Math. Soc. (2) **18** (1978), no. 2, 261–272.

[3] L. Carleson and N. G. Makarov, *Some results connected with Brennan's conjecture*, Israel J. Math. **87** (1994), 349–364.

[4] B. R. Choe, H. Koo, and W. Smith, *Compact difference of Fejér–Riesz composition operators*, Potential Anal. (2025).

[5] C. C. Cowen and B. D. MacCluer, *Composition Operators on Spaces of Analytic Functions*, Studies in Advanced Mathematics, CRC Press, Boca Raton, FL, 1995.

[6] P. L. Duren, *Univalent Functions*, Grundlehren der mathematischen Wissenschaften, vol. 259, Springer-Verlag, New York, 1983.

[7] H. Hedenmalm and S. Shimorin, *Weighted Bergman spaces and the integral means spectrum of conformal mappings*, Duke Math. J. **127** (2005), no. 2, 341–393.

[8] V. Matache and W. Smith, *Composition operators on Bergman spaces of simply connected domains*, Complex Var. Elliptic Equ. **57** (2012), no. 11, 1225–1238.

[9] OpenAI, *Brennan's conjecture and sharp inverse-square integral means*, Preprint (2026).

[10] Ch. Pommerenke, *On the integral means of the derivative of a univalent function*, J. London Math. Soc. (2) **32** (1985), 254–258.

[11] Ch. Pommerenke, *Boundary Behaviour of Conformal Maps*, Grundlehren der mathematischen Wissenschaften, vol. 299, Springer-Verlag, Berlin, 1992.

[12] J. H. Shapiro, *Composition Operators and Classical Function Theory*, Universitext: Tracts in Mathematics, Springer-Verlag, New York, 1993.

[13] W. Smith, *Composition operators on Bergman spaces of domains in the complex plane*, Complex Var. Elliptic Equ. **51** (2006), no. 8–11, 1081–1092.

[14] K. Zhu, *Operator Theory in Function Spaces*, 2nd ed., Mathematical Surveys and Monographs, vol. 138, American Mathematical Society, Providence, RI, 2007.